

# OCEANS AND CONTINENTS

Ocean basins are not just water-filled hollows in the Earth's surface. They are quite different from the continents that divide them. The heavy bedrock of ocean floors is the thin crust of the hot mantle that forms most of the planet, while continents are made of thicker slabs of lighter rock that float on the mantle like rafts. The continents are dragged slowly around the world by currents within the mantle. Oceans form where continents are pulled apart, and are destroyed where continents are pushed together.

## THE LAYERED EARTH ►

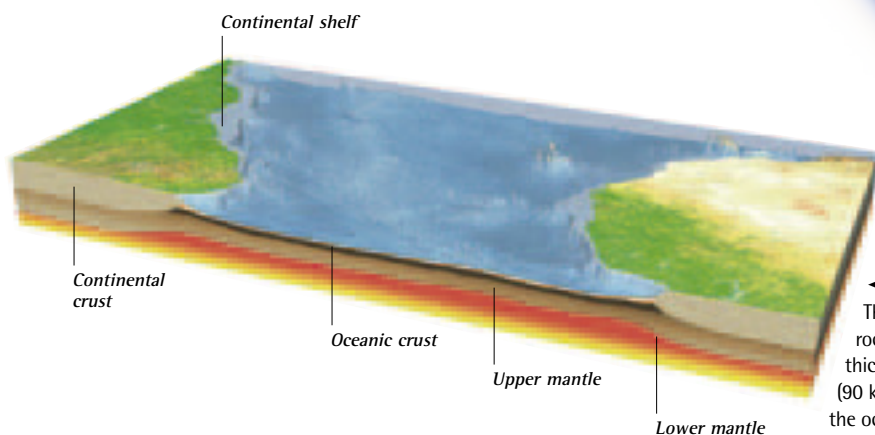
The Earth was formed from asteroids that were drawn together by gravity to form a rocky sphere. As they collided they released enough energy to melt the entire planet. Most of the heavy metal in the rock sank to the center to form its metallic core. The rock around the core became the solid but very hot mantle, and the skin of the mantle cooled to form the Earth's crust.

*Inner core: solid metals  
about 1,500 miles (2,440  
km) across; temperature  
12,600°F (7,000°C)*

*Outer core: liquid metals  
about 1,400 miles (2,250 km)  
thick; temperature 7,200°F  
(4,000°C)*

*Lower mantle: solid rock,  
1,400 miles (2,230 km)  
thick; temperature 6,300°F  
(3,500°C)*

*Upper mantle:  
partly molten  
rock, 416 miles (670  
km) thick; temperature  
1,800°F (1,000°C)*



## ◀ THE EARTH'S CRUST

The heavy, dark rock of the oceanic crust is similar to the rock of the upper mantle, and is just 4–7 miles (6–11 km) thick. By contrast, the continental crust is up to 55 miles (90 km) thick, but its rocks weigh less than the rocks that form the ocean floor. The continental shelves are the submerged edges of the continents, drowned by shallow coastal seas.

## MANTLE AND CRUST ROCKS

